

Lecture 2.2: The Rules of Falling

Unit 2: The Art of Observation

Today's Mission

- **Bridge Back:** Yesterday you learned to read the story of motion from graphs. The rocket's velocity-time graph showed constant acceleration during engine burn. Today, we'll develop the mathematical tools to predict exactly where and when objects reach their destinations.
- **Thinking Lens:** We will use **Mathematical Modeling** to create equations that describe constant acceleration motion.
- **Essential Questions:**
 - How do we derive the kinematic equations from first principles?
 - What makes freefall a special case of constant acceleration?
 - How can we systematically solve motion problems?

The Foundation: Constant Acceleration

When acceleration is constant, we can create precise mathematical relationships.

- **Definition of acceleration:**

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t}$$

- **Rearranging gives us our first kinematic equation:**

$$\vec{v}_f = \vec{v}_i + \vec{a}t \quad (1)$$

- This tells us the final velocity if we know initial velocity, acceleration, and time.

Deriving Position from Velocity

- For constant acceleration, the average velocity is:

$$\vec{v}_{avg} = \frac{\vec{v}_i + \vec{v}_f}{2}$$

- Since displacement equals average velocity times time:

$$\Delta\vec{x} = \vec{v}_{avg} \cdot t = \frac{\vec{v}_i + \vec{v}_f}{2} \cdot t$$

- Substituting equation (1), $\vec{v}_f = \vec{v}_i + \vec{a}t$:

$$\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2 \quad (2)$$

- This is our second kinematic equation.

The Time-Independent Equation

Sometimes we need to eliminate time from our calculations.

- Starting with:

- Equation (1): $\vec{v}_f = \vec{v}_i + \vec{a}t \rightarrow t = \frac{\vec{v}_f - \vec{v}_i}{\vec{a}}$

- Equation (2): $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$

- Substituting the expression for t :

$$\Delta\vec{x} = \vec{v}_i \left(\frac{\vec{v}_f - \vec{v}_i}{\vec{a}} \right) + \frac{1}{2}\vec{a} \left(\frac{\vec{v}_f - \vec{v}_i}{\vec{a}} \right)^2$$

- After algebraic manipulation:

$$\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x} \quad (3)$$

The Three Kinematic Equations

The Complete Set:

- Equation (1): $\vec{v}_f = \vec{v}_i + \vec{a}t$
- Equation (2): $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$
- Equation (3): $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}$

How to Choose:

- Each equation contains **four of the five variables**: $\vec{v}_i, \vec{v}_f, \vec{a}, t, \Delta\vec{x}$
- Select the equation that contains your **three known variables** and **one unknown**

Freefall: Galileo's Revolutionary Discovery

- **Key Insight:** All objects fall with the same acceleration (ignoring air resistance).
- On Earth's surface:

$$|\vec{a}_g| = 9.80 \text{ m/s}^2$$

- **Convention:** Choose positive direction as upward, so:

$$\vec{a} = -\vec{a}_g = -9.80 \text{ m/s}^2$$

- Freefall is simply constant acceleration motion with $\vec{a} = -9.80 \text{ m/s}^2$!
- **Watch:** [Galileo's Discovery of Freefall](#)

Problem-Solving Strategy

Steps 1-3:

- **Step 1:** Identify what you know and what you want to find
- **Step 2:** Choose coordinate system (positive direction)
- **Step 3:** Select the kinematic equation that contains your known and unknown variables

Steps 4-6:

- **Step 4:** Substitute values with proper units and significant figures
- **Step 5:** Solve algebraically, then substitute numerical values
- **Step 6:** Check your answer for reasonableness

Worked Example: Stone Drop

Problem: A stone is dropped from rest from a height of 45.0 m. How long does it take to reach the ground, and what is its final velocity?

- **Given:**
 - $\vec{v}_i = 0.0 \text{ m/s}$ (dropped from rest)
 - $\Delta\vec{x} = -45.0 \text{ m}$ (downward displacement)
 - $\vec{a} = -9.80 \text{ m/s}^2$
- **Find:** t and $\vec{v}_f$

Stone Drop Solution: Finding Time

Use equation (2): $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$

- Since the stone is dropped from rest, $\vec{v}_i = 0.0 \text{ m/s}$:

$$\Delta\vec{x} = \frac{1}{2}\vec{a}t^2$$

- Solve algebraically for t :

$$t = \sqrt{\frac{2\Delta\vec{x}}{\vec{a}}}$$

- Now substitute numerical values:

$$t = \sqrt{\frac{2(-45.0 \text{ m})}{-9.80 \text{ m/s}^2}} = \sqrt{9.18 \text{ s}^2} = 3.03 \text{ s}$$

Stone Drop Solution: Finding Velocity

Use equation (1): $\vec{v}_f = \vec{v}_i + \vec{a}t$

- Solve algebraically (with $\vec{v}_i = 0$):

$$\vec{v}_f = \vec{a}t$$

- Substitute numerical values:

$$\vec{v}_f = (-9.80 \text{ m/s}^2)(3.03 \text{ s}) = -29.7 \text{ m/s}$$

The negative sign indicates downward motion.

- **Watch:** [Falling Objects Without Air Resistance](#)

Worked Example: Rocket at Apogee

Problem: A model rocket is launched vertically upward with initial velocity $\vec{v}_i = 125 \text{ m/s}$. Find the maximum height and time to reach it.

- **Given:**
 - $\vec{v}_i = 125 \text{ m/s}$ (upward)
 - $\vec{v}_f = 0.0 \text{ m/s}$ (at maximum height)
 - $\vec{a} = -9.80 \text{ m/s}^2$ (gravitational acceleration)
- **Find:** t and $\Delta\vec{x}$

Rocket Solution: Finding Time

Use equation (1): $\vec{v}_f = \vec{v}_i + \vec{a}t$

- Solve algebraically for t :

$$t = \frac{\vec{v}_f - \vec{v}_i}{\vec{a}}$$

- Substitute numerical values:

$$t = \frac{0.0 \text{ m/s} - 125 \text{ m/s}}{-9.80 \text{ m/s}^2} = 12.8 \text{ s}$$

Rocket Solution: Finding Height

Use equation (3): $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}$

- Solve algebraically for $\Delta\vec{x}$:

$$\Delta\vec{x} = \frac{\vec{v}_f^2 - \vec{v}_i^2}{2\vec{a}}$$

- Substitute numerical values:

$$\Delta\vec{x} = \frac{(0.0 \text{ m/s})^2 - (125 \text{ m/s})^2}{2(-9.80 \text{ m/s}^2)} = 797 \text{ m}$$

The rocket reaches a maximum height of 797 m.

Check for Understanding

A ball is thrown upward with initial velocity 15.0 m/s.

1. Which equation would you use to find how high it goes?

- Use equation (3), $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}$ with $\vec{v}_f = 0.0 \text{ m/s}$ at maximum height
- Solve algebraically: $\Delta\vec{x} = \frac{\vec{v}_f^2 - \vec{v}_i^2}{2\vec{a}}$
- Substitute values: $\Delta\vec{x} = \frac{(0.0 \text{ m/s})^2 - (15.0 \text{ m/s})^2}{2(-9.80 \text{ m/s}^2)} = \frac{-225 (\text{m/s})^2}{-19.6 \text{ m/s}^2} = 11.5 \text{ m}$

Check for Understanding (continued)

2. What is the acceleration throughout the motion?

- $\vec{a} = -9.80 \text{ m/s}^2$ (constant throughout entire flight)

3. How long does it take to return to the starting point?

- First find time to reach maximum height: $t_{up} = \frac{\vec{v}_f - \vec{v}_i}{\vec{a}} = \frac{0.0 \text{ m/s} - \vec{v}_i}{\vec{a}}$
- By symmetry, total flight time = $2t_{up} = \frac{2(15.0 \text{ m/s})}{9.80 \text{ m/s}^2} = 3.06 \text{ s}$

Connection to Our Rocket Investigation

The Power of Equations:

- We can now predict when the rocket engine cuts off
- We can calculate exactly when it reaches maximum altitude
- We can determine its velocity at any point in the trajectory
- These same equations govern satellites, projectiles, and all falling objects

From graphs to equations: Yesterday's slopes and areas are now precise mathematical relationships!

Answering Our Essential Questions

1. How do we derive the kinematic equations from first principles?

- Start with definition of acceleration: $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$
- Apply algebraic manipulation to eliminate variables

2. What makes freefall a special case of constant acceleration?

- Freefall has constant acceleration $\vec{a}_g = -9.80 \text{ m/s}^2$
- Same kinematic equations apply with this specific acceleration value (on Earth)

3. How can we systematically solve motion problems?

- Identify knowns and unknowns • Choose appropriate equation • Solve algebraically first

Summary and Next Steps

The Kinematic Toolbox:

- Equation (1): $\vec{v}_f = \vec{v}_i + \vec{a}t$
- Equation (2): $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$
- Equation (3): $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}$

Key Principles:

- Always define coordinate system clearly
- Use proper vector notation and units throughout
- Freefall is constant acceleration with $\vec{a} = -9.80 \text{ m/s}^2$